

Thrishantha Nanayakkara

Senior Lecturer

EDUCATION

- Sept 1998 - August 2001 **PhD in Systems Control and Robotics**, Department of Mechanical Engineering, Saga University, Saga, Japan
- PhD Thesis title *An Evolutionary Approach to Skillful Motion Control of Industrial Robots*
- adviser Prof. Keigo Watanabe, Department of Mechanical Engineering, Saga University, 1-Honjomachi, Saga 840-8502, Japan
- description The main focus of the thesis was to develop biologically inspired computational algorithms to control position and force of robot manipulators with redundant degrees of freedom, and to make them learn through interactions with the environment. It was proved that nonlinear model primitives learned in partitions of the workspace can be adaptively combined to control an industrial robot arm with 7-degrees of freedom along desired velocity and force profiles. Findings were published in 2-invited book chapters, 5-refereed international journals, and 16 international conferences [10]-[12], [14], [16]-[18], [46]-[61]
- Sept 1996 - August 1998 **MSc in Electrical Engineering**, Department of Electrical Engineering, Saga University, Saga, Japan
- MSc Thesis title *Accurate Contour Control of Industrial Robot Manipulators*
- adviser Prof. Masatoshi Nakamura, Department of Electrical Engineering, Saga University, 1-Honjomachi, Saga 840-8502, Japan
- description The focus was to develop a method to control industrial articulated robot manipulators in contour control tasks (arc welding, laser cutting, grinding etc.). Experiments were done using a Yasakawa robot with 4-degrees of freedom. Results were presented at the Japanese Society of Instrument and Control Engineers conference, 1997 [62]
- April 1992 - August 1996 **BSc in Electrical Engineering, 1st class honors**, Department of Electrical Engineering, University of Moratuwa, Moratuwa, Sri Lanka
- description University of Moratuwa, Sri Lanka accepts the top 0.2% of the students who sit for the GCE(A/L) examination. I carried out two projects: one official and one voluntary. The official project was on steady state analysis of grid connected induction generators. This project won the best undergraduate project award of the Electrical Engineers Society, University of Morwtawa, in 1996. In addition, I carried out a voluntary project in collaboration with the Kyushu Electric Company, Japan, on intelligent optimization of burning conditions of a 700MW coal fired power plant to achieve desired chemical properties of fly-ash (to reduce the cost of processing fly-ash to produce paint, cosmetics, tar, etc.). The results were presented at the International Federation of Automatic Control - Systems Identification conference, 1996 [63].
- April 1990 - April 1992 **Passed finalist of the Chartered Institute of Management Accountants-UK**, Colombo Business School, Colombo Sri Lanka

WORK EXPERIENCE

- September 2012 - present **Senior Lecturer**, Division of Engineering, King's College London, London, United Kingdom
- March 2008 - September 2012 **Lecturer**, Division of Engineering, King's College London, London, United Kingdom

July 2007 - January 2008	Postdoctoral research fellow , <i>School of Engineering and Applied Sciences</i> , Harvard University, Cambridge, MA. United States
February 2008 - August 2008	Visiting Scholar , <i>School of Engineering and Applied Sciences</i> , Harvard University, Cambridge, MA. United States
September 2008 - June 2009	Radcliffe Fellow , <i>Radcliffe Institute for Advanced Studies</i> , Harvard University, Cambridge, MA. United States
April 2008 - June 2009	Research Affiliate , <i>Computer Science and Artificial Intelligence Laboratory (CSAIL)</i> , MIT, Cambridge, MA. United States
description	My research focus during the stay at Harvard University and MIT was in stochasticity of legged locomotion, the role of passive degrees of freedom in locomotion in unstructured environments, stochasticity of odor guided behavior of rodents, and robotic learning in animal-robot mixed colonies. Findings were published in 2 invited book chapters [8], [9] and 1 international Workshop organized by the International Advanced Robotics Programme [67]. One paper was published in Neuron.
August 2003 - June 2008	Senior lecturer , <i>Department of Mechanical Engineering</i> , University of Moratuwa, Moratuwa. Sri Lanka
description	I initiated a research program on animal-robot mixed colonies for odor localization in unstructured environments. The work has been featured in New Scientist Magazine (issue 2653, 26 April 2008), Yorkshire Post (Sunday, 11th May 2008), and the Harvard Magazine (Sept.-October issue, 2008) on animal-robot integrated systems to detect landmines. The Engineering contribution to the 2004 Tsunami relief projects done by his team at the University of Moratuwa appeared in the Chronicle of Higher Education on 21st January, 2005.
November 2001 - July 2003	Postdoctoral Fellow , <i>Department of Biomedical Engineering, School of Medicine</i> , Johns Hopkins University, Baltimore, MD. United States
description	My research focus was to understand how the human motor system learns internal models of hand and task dynamics to stabilize and control the human hand along desired velocity and force profiles. I conducted controlled experiments on human subjects to study the adaptive properties of these internal models when the task dynamics change over time. Findings were published in the Journal of Neuroscience and the Journal of Neurophysiology [15], [13].

AWARDS AND HONORS

2013	Nominated to IEEE Franklin Taylor Best Paper award , co-author of the paper titled "An Optimal State Dependent Haptic Guidance Controller Via a Hard Rein", <i>IEEE International Conference on Systems man and cybernetics</i> , SMC-2013, October, Manchester. UK
2012	Best paper award , co-author of the paper titled "A Computationally Efficient Framework for Stochastic Prediction of Flood Propagation", <i>IEEE International Conference on Information and Automation for Sustainability</i> , ICIAfS - 2012, September 27-29, Beijing. China
2012	2012 Royal Academy of Engineering ERA Foundation Entrepreneurs Award finalist , Royal Academy of Engineering, London. UK
2011	Innovations Fellow , King's College, London, London. UK
2008	Radcliffe Fellowship , Harvard University, Cambridge, MA. United States
description	This fellowship was awarded by nomination by faculty members of the Harvard University based on the individual's academic merit and potential to serve pressing needs of the society
2005	Outstanding Research Personnel award , University of Moratuwa, Moratuwa. Sri Lanka
description	Awarded based on peer reviewed publications
2001-2003	Postdoctoral Research Fellowship , <i>Department of Biomedical Engineering</i> , Johns Hopkins University, Baltimore, MD. United States

description	Awarded to carry out independent research on adaptive predictive controllers in the human motor system
2000-2001	Saga Prefecture scholarship for postgraduate studies , Saga. Japan
description	Based on the recommendation of Saga University
1996	Monbusho Fulbright Scholarship for postgraduate studies , Ministry of Education. Japan
description	Based on the recommendation of Saga University
1996	Award for the best undergraduate research project , <i>Electrical Engineering Society</i> , Department of Electrical Engineering, University of Moratuwa. Sri Lanka
description	Based on demonstration, oral presentation, and viva
1989	Mahapola Fulbright Scholarship , Ministry of Education. Sri Lanka
description	awarded by the Government of Sri Lanka for the top 10% of the students to undergo undergraduate education in Sri Lanka

RESEARCH GRANTS

January, 2012-Dec, 2015	European Union FP7 Call-7 grant , <i>Call identifier: FP7-ICT-2009-7</i> , Co-Investigator, technical manager of the consortium, (KCL amount (coordinator): €1.3m, total consortium grant: €7.38m). UK
Title	STIFF-FLOP - STIFFness controllable Flexible and Learn-able Manipulator for surgical Operations
March, 2011-Sept, 2012	EPSRC first grant , <i>Engineering and Physical Sciences Research Council</i> , Principal Investigator, (amount: £97k). UK
Title	Impedance control on uncertain objects
April, 2011-Sept, 2014	EPSRC , <i>Engineering and Physical Sciences Research Council</i> , Co-Investigator in collaboration with Sheffield Hallam University, (amount: £200k). UK
Title	REINS: Human robot interaction through reins
March, 2011-Dec, 2014	European Union FP7 Call-6 grant , <i>Call identifier: FP7-ICT-2009-6</i> , Co-Investigator, (amount: £535k). UK
Title	TOMSY-Topological Motion Synthesis for Dexterous Manipulation
March, 2011-Dec, 2014	European Union FP7 Call-6 grant , <i>Call identifier: FP7-ICT-2009-6</i> , Co-Investigator, (amount: €480k). UK
Title	DARWIN-Dextrous Assembler Robot Working with embodied Intelligence (I have been appointed as the KCL coordinator/PI in the absence of Prof. Lakmal Seneviratne)
Sept, 2008-April, 2009	Award to organize an exploratory seminar , <i>Radcliffe Institute for Advance Studies, Harvard University</i> , Co-organizer with Prof. Robert Howe, Associate Dean, School of Engineering and Applied Sciences, Harvard University, (amount: USD14k). USA
Title	Radcliffe Exploratory Seminar on Technologies for Humanitarian Demining
Jan, 2006-Dec, 2006	Grant no. RG/ 2006/TDPS/02 , <i>National Science Foundation of Sri Lanka (NSF)</i> , Principal Investigator, (amount: Rs. 2,720k). Sri Lanka
Title	Deployment of a colony legged mobile robots for landmine detection: An extension to MURALI I (Moratuwa University Robot for Anti- landmine Intelligence)
Jan, 2005-Dec, 2006	Grant no. 12/620/467 , <i>Council for Agricultural Research Policy of Sri Lanka (CARP)</i> , Principal Investigator, (amount: Rs. 553.12k). Sri Lanka
Title	Development of an embedded sensor to detect red-weevil larvae in early stages
Jan, 2004-Dec, 2005	Grant no. RG/ 2004/ E02 , <i>National Science Foundation</i> , Principal Investigator, (amount: Rs. 902.5k). Sri Lanka

Title A robotic system for landmine detection

TEACHING EXPERIENCE

- 2012-2013 **5CCS2ARS**, *Adaptive and Robotic Systems*, for 2nd year undergraduate students, King's College, University of London.
United Kingdom
- 2009-2013 **6CCS3PRJ/7CCS4PRJ**, *Real time systems and Control*, for 3rd year undergraduate and masters students, King's College, University of London.
United Kingdom
- 2009-2012 **5CEEE504**, *Computer Engineering II - Computer Aided Design part (eight lectures)*, for 2nd year undergraduate students, King's College, University of London.
United Kingdom
- 2004-2006 **ME428**, *Automation and Robotics*, for 4th year undergraduate students, University of Moratuwa.
Sri Lanka
- 2004-2006 **ME301**, *Control Systems and Instrumentation*, for 3rd year undergraduate students, University of Moratuwa.
Sri Lanka
- 2003-2004 **EE222**, *Electrical Measurements*, for 2nd year undergraduate students, University of Moratuwa.
Sri Lanka
- 2003-2004 **CS323**, *Intelligent Systems*, for 3rd year undergraduate students, University of Moratuwa.
Sri Lanka
- 2003-2007 **EE5061**, *State Space Design*, for MSc in industrial Automation, University of Moratuwa.
Sri Lanka
- 2003-2007 **ME5144**, *Mechatronics and Robotics*, for MSc in industrial Automation, University of Moratuwa.
Sri Lanka

GUEST LECTURES

- May 31st 2011 **Legged locomotion on soft terrains**, *bridging the autonomous divide & knowledge transfer network seminar*, Aerospace, Aviation, and Defence network.
UK
- August 3rd 2010 **The Choice of Action and the Statistical Properties of Motor Primitives**, *UK-China Research Seminar, Bioengineering Group, School of Mechanical, Aerospace and Civil Engineering (MACE)*, University of Manchester.
UK
- April 8th 2008 **Humanitarian demining, challenges and future research directions**, *Mechanics: Modeling, Experimentation & Computation seminar series (MMEC)*, Computer Science and Artificial Intelligence Laboratory, MIT.
United States
- February 5th 2008 **Animal Robot Mixed Colonies for Humanitarian Demining**, *Center for Research on Computing for Society*, Harvard University.
United States
- December 7th 2007 **Animal Robot Mixed Colonies for Humanitarian Demining**, *Department of Mechanical Engineering*, University of Melbourne.
Australia
- March 29th 2007 **Sustainable Technology for Disaster Response**, *TechBridgeWorld and Field Robotics Center Joint Seminar*, Robotics Institute, Carnegie Mellon University.
United States

INVITED PROFESSIONAL APPOINTMENTS

- 2013 **Member, Program Committee**, *World Automation Congress*, 2014.
Hawaii
- 2013 **Founding member**, *London Robotics Network*, Knowledge Transfer Network.
United Kingdom

- 2013 **Panel member**, *Large research grants*, Engineering and Physical Sciences Research Council.
United Kingdom
- 2013 **Member, Program Committee**, *IEEE International Conference on Mechatronics and Automation*, 2013.
Japan
- 2012 **EPSRC college**, *Invited to be a member of the EPSRC college of peer reviewers*, From Sept. 2012.
UK
- 2012 **Co-chair**, *the session on Design of Medical Robots (Regular Session), Track T9, in the IEEE/RSJ International Conference on Intelligent Robots and Systems*, Tuesday, October 9, 2012, Vilamoura.
Portugal
- 2012 **Program committee**, *2012 IEEE/SICE International Conference on System Integration (SII2012)*, December 16-18, 2012, Fukuoka.
Japan
- 2012 **Program committee**, *2012 IEEE International Conference on Mechatronics and Automation (IEEE ICMA 2012)*, August 5 to August 8, 2012, Sichuan.
China
- 2012 **Program committee**, *IECBES 2012 (2012 IEEE EMBS International Conference on Biomedical Engineering and Sciences)*, December 17, 2012 - Dec 19, 2012, Langkawi.
Malaysia
- 2012 **Publications Chair**, *IEEE International Conference on Information and Automation for Sustainability*, ICIAfS - 2012, September 27-29, Beijing.
China
- 2010 **Publications Chair**, *IEEE International Conference on Information and Automation for Sustainability*, ICIAfS - 2010, December 17-19, Colombo.
Sri Lanka
- 2009 **Panel member**, *Research grants*, Engineering and Physical Sciences Research Council.
United Kingdom
- 2009-now **Reviewer**, *Research grants*, Engineering and Physical Sciences Research Council.
United Kingdom
- 2005-now **Associate editor**, *Journal of Control and Intelligent Systems*, editor in chief: Prof. Clarence de Silva, ACTA Press.
Canada
- 2007-now **PhD Thesis reviewer**, *University of Melbourne*.
Australia
- 2007-now **PhD Thesis reviewer**, *University of Deakin*.
Australia
- 2008-now **Guest editor**, *special issue on Humanitarian Technologies*, *Journal of Control and Intelligent Systems*, editor in chief: Prof. Clarence de Silva, ACTA Press.
Canada
- 2008 **Publications co-chair**, *IEEE/ACM International Conference on Information and Communication Technologies and Development (ICTD - 2009)*, April 17 - 19, Carnegie Mellon Qatar, Doha.
Qatar
- 2008 **General Co-Chair**, *International Conference on Information and Automation for Sustainability (ICIAfS - 2008)*, Dec 12-14, Colombo.
Sri Lanka
- 2007 **Program chair**, *International Conference on Information and Automation for Sustainability (ICIAfS - 2007)*, Dec 3-6, Melbourne.
Australia
- 2006 **Secretary**, *Sri Lanka Chapter*, Institute of Electrical and Electronics Engineers (IEEE).
Sri Lanka
- 2005-2007 **Commissioner**, *Sri Lanka Inventors Commission*, Ministry of Investment Promotion and Enterprise Development. (Task involved promoting innovative projects in 6000 secondary schools and coordinate product incubation projects in a selected number of Universities in Sri Lanka. Handled an annual budget of USD1.5 million).
Sri Lanka

- 2006-2007 **Steering Committee member**, *Centre for Research and Development*, Ministry of Defence.
Sri Lanka
- 2006-2007 **Member, Board of Governors**, *Arthur C. Clarke Institute for Modern Technology*, Ministry of Science and Technology.
Sri Lanka
- 2005 **Committee member**, *Ministerial committee to develop an evaluation criteria to reward scientists*, Ministry of Science and Technology.
Sri Lanka
- 2005 **Committee member**, *Ministerial scoping committee to revise the curriculum for the General Certificate of Education (GCE) Ordinary Level (O/L) examination*, Ministry of Education.
Sri Lanka
- 2005 **Founding general chair**, *International Conference on Information and Automation, Dec 16-18, Colombo*, (with technical co-sponsorship of IEEE region 10, IEEE SMC society).
Sri Lanka
- 2005 **President**, *Sri Lanka Association for Artificial Intelligence*.
Sri Lanka

INTERNATIONAL NEWS AND FEATURE ARTICLES

- October issue, 2012 **IEEE Spectrum magazine**, *Robotics news on Inflatable Limb Robot Runs Around on Wiggly Legs*.
USA
- September issue, 2008 **Harvard Magazine**, *On animal-robot interaction studies*.
USA
- Feb 17, 2009, and April 29, 2009 **Harvard Crimson**, *On technology and human rights*.
USA
- Issue 2653, 26th April 2008 **New Scientist Magazine**, *On legged robots and animal odor localization*.
UK
- 11th May, 2008 **Yorkshire Post**, *On animal-robot interaction studies*.
UK

Publications

- [1] Taylor, John G., Vassilis Cutsuridis, Matthew Hartley, Kaspar Althoefer, and Thrishantha Nanayakkara. "Observational Learning: Basis, Experimental Results and Models, and Implications for Robotics." *Cognitive Computation*: 1-15.
- [2] Intelligent Control Systems with an Introduction to System of Systems Engineering by Thrishantha Nanayakkara, Ferat Sahin, and Mo Jamshidi (Hardcover - 15 Oct 2009), publisher: CRC Press, Taylor and Francis group
- [3] Xiaojing Song, Hongbin Liu, Kaspar Althoefer, Thrishantha Nanayakkara, and Lakmal Seneviratne, "Efficient Break-Away Friction Ratio Prediction Based on Haptic Surface Exploration", (in press), *IEEE Transactions on Robotics*, 2012.
- [4] Wijesundera, Isuri, Malka N. Halgamuge, Thas Nirmalathas, and Thrishantha Nanayakkara. "A Geographic Primitive-Based Bayesian Framework to Predict Cyclone-Induced Flooding*." *Journal of Hydrometeorology* 14, no. 2 (2013): 505-523.
- [5] Thrishantha Nanayakkara, Malka N. Halgamuge, Prasanna Sridhar, and Asad M. Madni, "Intelligent Sensing in Dynamic Environments Using Markov Decision Process", *Sensors*, vol. 11, no. 1, pp. 1229-1242, 2011.
- [6] Gary Chin-Wei Sing, Wilsan M Joiner, Thrishantha Nanayakkara, Jordan B Brayanov, and Maurice Smith, "Primitives for motor adaptation reflect correlated neural turning to position and velocity", *Neuron*, vol. 64, Issue 4, pp. 575-589, November 2009.
- [7] K.A.P. Siriwardena, L.C.P. Fernando, N. Nanayakkara, K. F. G. Perera, A.D.N.T. Kumara and T. Nanayakkara, "Portable Acoustic Device for Detection of Coconut Palms infested by *Rynchophorus ferrugineus* (Coleoptera: Curculionidae)", *Journal of Crop Protection*, vol. 29, pp. 25-29, 2010
- [8] Thrishantha Nanayakkara, Tharindu Dissanayake, Prasanna Mahipala and K. A. Gayan Sanjaya (2008). *A Human-Animal-Robot Cooperative System for Anti-Personal Mine Detection*, Humanitarian Demining, (chapter 15), pp. 347 - 366, Maki K. Habib (Ed.), ISBN: 978-3-902613-11-0, I-Tech Education and Publishing,

- [9] Thrishantha Nanayakkara, Lasitha Piyathilaka, and Akila Subasingha "Mechatronics in Landmine Detection and Removal", MECHATRONIC SYSTEMS Devices, Design, Control, Operation, and Monitoring, Edited by Clarence De Silva, CRC Press, Taylor and Francis, Boca Raton, FL Chapter 28, 2007.
- [10] K. Watanabe, Thrishantha Nanayakkara, Kazuo Kiguchi, and Kiyotaka Izumi "Achieving synergy through acquisition of human skill," A New Life-Style in 21 Century Living with Cognitive and Behavioral Intelligent Artificial Liferobot, Edited by Masanori Sugisaka, Springer-Tokyo, Chapter 40, 2005.
- [11] Thrishantha Nanayakkara, Keigo Watanabe, Kazuo Kiguchi, and Kiyotaka Izumi, "Evolutionary Dynamics Identification of Multi-Link Manipulators Using Runge–Kutta–Gill RBF Networks," Soft Computing in Measurement and Information Acquisition, Edited by L. Reznik and V. Kreinovich, Physica-Verlag, pp.208–222, 2003.
- [12] Thrishantha Nanayakkara, Keigo Watanabe, Kazuo Kiguchi and Kiyotaka Izumi, "Evolving a multi-objective obstacle avoidance skill of a seven-link manipulator subject to constraints," International Journal of Systems Science, vol. 35 , no. 3, pp. 167 - 177, March 2004.
- [13] Thrishantha Nanayakkara and Reza Shadmehr, "Saccade Adaptation in Response to Altered Arm Dynamics", Journal of Neurophysiology, 2003, 90:4016-4021.
- [14] Kazuo Kiguchi, Thrishantha Nanayakkara, Keigo Watanabe, Toshio Fukuda, "Multi-Dimensional Reinforcement Learning Using a Vector Q-Net - Application to Mobile Robots", International Journal of Control, Automation and Systems, vol.1, no.1, pp.142-148, 2003.
- [15] Gregory Ariff, Opher Donchin, Thrishantha Nanayakkara, and Reza Shadmehr, "A Real-Time State Predictor in Motor Control: Study of Saccadic Eye Movements During Unseen Reaching Movements," Journal of Neuroscience, vol. 22, no. 17, pp. 7721-7729, Sept. 2002.
- [16] Thrishantha Nanayakkara, Keigo Watanabe, Kazuo Kiguchi and Kiyotaka Izumi, "Evolutionary Learning of a Fuzzy behavior Based Controller for a Non-Holonomic Mobile Robot in a Class of Dynamic Environments," International Journal of Intelligent and Robotic Systems, vol.32, no.3, pp.255-277, November, 2002.
- [17] Thrishantha Nanayakkara, Kazuo Kiguchi, Tsukasa Murakami, Keigo Watanabe, Kiyotaka Izumi, "Enhancing the Autonomy of Teleoperated Redundant Manipulators Through Fusion of Intelligent Control Modules," International Journal of Robotics and Mechatronics, vol.14, no.3, pp.534-545, June, 2002.
- [18] Thrishantha Nanayakkara, Keigo Watanabe, Kazuo Kiguchi and Kiyotaka Izumi, "Fuzzy Self-Adaptive RBF Neural Network Based Control of a Seven-Link Industrial Robot Manipulator," in Journal of Advanced Robotics, vol. 15, no. 1, pp. 17-43, 2001.
- [19] Milad S. Malekzadeh, Danilo Bruno, Sylvain Calinon, Thrishantha Nanayakkara, Darwin G. Caldwell, "Skills Transfer across Dissimilar Robots by Learning Context-Dependent Reward" accepted in IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2013.
- [20] Anuradha Ranasinghe, Prokar Dasgupta, Jaques Penders, Kaspar Althoefer, Thrishantha Nanayakkara, "A Two Party Haptic Guidance Controller Via a Hard Reinforcement", accepted in IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2013.
- [21] Valerio Pereno, Kya Shoar, Giulia Bartoli, Fabio Bianchi, Thrishantha Nanayakkara, "Stable Walking on Variable Visco-Elastic Terrains using Meta-parameters for Passive State Migration", accepted in IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2013.
- [22] J. Konstantinova, M. Li, V. Aminzadeh, P. Dasgupta, K. Althoefer, and T. Nanayakkara, "Force-Velocity Modulation Strategies for Soft Tissue Examination, (Periodical style)" Accepted for publication", accepted in IEEE International conference on intelligent Robots and systems (IROS), 2013.
- [23] Miguel González-Fierro, Carlos Balaguer, Nicola Swann, Thrishantha Nanayakkara, "A Humanoid Robot Standing Up Through Learning from Demonstration Using a Multimodal Reward Function", IEEE-RAS International Conference on Humanoid Robots 2013.
- [24] Anuradha Ranasinghe, Prokar Dasgupta, Jaques Penders, Kaspar Althoefer, Thrishantha Nanayakkara, "An Optimal State Dependent Haptic Guidance Controller Via a Hard Reinforcement", Accepted to publish IEEE International Conference on system, man, cybernetics (SMC) 2013.
- [25] A. Jiang, T. Aste, P. Dasgupta, K. Althoefer, and T. Nanayakkara, "Granular jamming with hydraulic control," ASME 2013 International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC 2013), Portland, OR, USA, 2013.
- [26] A. Jiang, T. Aste, P. Dasgupta, K. Althoefer, and T. Nanayakkara, "Granular jamming transitions for a robotic mechanism," Powders & Grains 2013, Sydney, Australia, 2013.
- [27] A. Jiang, P. Agrawal, K. Althoefer, T. Nanayakkara, P. Dasgupta, , "Bio-inspired connective granular jamming for a robotic limb," 35th International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC 2013), Osaka, Japan, 2013.

- [28] A. Jiang, K. Althoefer, P. Dasgupta, T. Nanayakkara, "The Core-Snake, the variable stiffness laparoscopic camera," The Hamlyn Symposium on Medical Robotics, London, United Kingdom, 2013.
- [29] Masinghe, Waruna, Gordana Collier, Andrzej Ordys, and Thrishantha Nanayakkara. "A novel approach to determine the inverse kinematics of a human upper limb model with 9 degrees of freedom." In Biomedical Engineering and Sciences (IECBES), 2012 IEEE EMBS Conference on, pp. 525-530. IEEE, 2012.
- [30] Isuru S. Godage, Thrishantha Nanayakkara, and Darwin G. Caldwell, "Locomotion with Continuum Limbs", in IEEE/RSJ International Conference on Intelligent Robots and Systems, pp. 293 - 298, October 7-12, 2012 Vilamoura, Algarve, Portugal.
- [31] Allen Jiang, Joao Bimbo, Simon Goulder, Hongbin Liu, Xiaojin Song, Prokar Dasgupta, Kaspar Althoefer, and Thrishantha Nanayakkara, "Adaptive grip control on an uncertain object", IEEE/RSJ International Conference on Intelligent Robots and Systems, pp. 1161 - 1166, October 7-12, 2012 Vilamoura, Algarve, Portugal.
- [32] Allen Jiang, G. Xynogalas, Prokar Dasgupta, Kasper Althoefer, and Thrishantha Nanayakkara, "Design of a Variable Stiffness Flexible Manipulator with Composite Granular Jamming and Membrane Coupling", IEEE/RSJ International Conference on Intelligent Robots and Systems, pp. 2922 - 2927, October 7-12, 2012 Vilamoura, Algarve, Portugal.
- [33] Fabio Bianchi, Giulia Bartoli, Kya Shoar, Maria R. Armas Fernandez, Valereo Pereno, Jelizaveta Zirkova, Allen Jiang and Thrishantha Nanayakkara, "Adaptive Internal Impedance Control for Stable Walking on Uncertain Visco-elastic Terrains", IEEE/RSJ International Conference on Intelligent Robots and Systems, pp. 2465 - 2470, October 7-12, 2012 Vilamoura, Algarve, Portugal.
- [34] Thrishantha Nanayakkara, Katie Byl, Hongbin Liu, Xiaojing Song, Tim Villabona, "Dominant Sources of Variability in Passive Walking", IEEE International Conference on Robotics and Automation, pp. 1003-1010, RiverCentre, Saint Paul, Minnesota, USA, pp. 1003 - 1010, May 14-18, 2012 (40% acceptance rate).
- [35] Hongbin Liu, Xiaojing Song, Thrishantha Nanayakkara, Lakmal Seneviratne, Kaspar Althoefer, "Real-Time Local Contact Shape and Pose Classification using a Tactile Array Sensor", IEEE International Conference on Robotics and Automation, pp. 1410 - 1415, RiverCentre, Saint Paul, Minnesota, USA, May 14-18, 2012 (40% acceptance rate).
- [36] Jonathan Lipinette, Matthew Valente, and Thrishantha Nanayakkara, "Careful Walking on Soft Terrains with Deformable Robots", HUDEM 2011, Croatia, August 2011.
- [37] Prasanna Sridhar, Thrishantha Nanayakkara, Asad M. Madni, and Mo Jamshidi, "Dynamic Power Management of an Embedded Sensor Network based on Actor-Critic Reinforcement based Learning", in Procs. of the 3rd International Conference on Information and Automation for Sustainability (IEEE technical co-sponsor), pp. 76 - 81, Dec. 4 - 8, Melbourne, Australia, 2007.
- [38] Thrishantha Nanayakkara, Lasitha Piyathilaka, Akila Subasingha, and Mo Jamshidi, "Development of Advanced Motor Skills in a Group of Humans through an Elitist Visual Feedback Mechanism", in Procs. of IEEE International Conference on Systems of Systems Engineering, San Antonio, 2007.
- [39] Thrishantha Nanayakkara, Lanka Udawatta, Anjula De Silva, Madura Jayathilake, Sanjaka Peiris, Asanke Galgomuwa, "High Performance Temperature Controller for Infant Incubators " in Proc. of the International Conference on Information and Automation 2006, Colombo , Sri Lanka pp. 115-120, 2006
- [40] Thrishantha Nanayakkara, Lasitha Piyathilake, and Akila Subasinghe, "A Simplified Statistical Approach to Classify Vegetated Environments for Robotic Navigation in Tropical Minefields", Proc. of the International Conference on Information and Automation (IEEE technical co-sponsor), pp. 337 - 342, Colombo, Sri Lanka, Dec. 15-18, 2005
- [41] Lakmal Silva and Thrishantha Nanayakkara, "Impedance Control of a Robot Manipulator for Demining", Procs. of the International Conference on Information and Automation (IEEE technical co-sponsor), pp. 282 - 287, Colombo, Sri Lanka, Dec. 15-18, 2005
- [42] M. Bernardine Dias, G. Ayorkor Mills-Tettey, and Thrishantha Nanayakkara, "Robotics, Education, and Sustainable Development", in Proc. of IEEE International Conference on Robotics and Automation (ICRA2005), pp. 4259 - 4264, Barcelona, Spain, 2005
- [43] V. Kumarathasan and Thrishantha Nanayakkara, " Intelligent Collaboration Among Robotic Agents for Landmine Detection", Proc. of the Annual Sessions, Sri Lanka Association for Artificial Intelligence, 2005.
- [44] V Aravinthan, Thrishantha Nanayakkara, I J Dayawansa, "Adaptive Resonance Theory based Neural Network approach for Signal distribution in VLF metal detectors", in Procs. of the IEE Annual Conference 2004, Sri Lanka Section.

- [45] Ampikathan Aravinthan and Thrishantha Nanayakkara, "Implementing Behavior Based Control in an Autonomous Mine Detecting Robot (AMDR)", Engineering Research Unit Symposium 2004, University of Moratuwa.
- [46] Thrishantha Nanayakkara, Keigo Watanabe, Kazuo Kiguchi, Reza Shadmehr, "Intelligent Control of a 7-DOF manipulator based on model primitives", in Proc. of IEEE International Symposium on Intelligent Control (ISIC2002), pp. 538 - 543, 2002, Vancouver, Canada.
- [47] Kazuo Kiguchi, Thrishantha Nanayakkara, Keigo Watanabe, Toshio Fukuda, "Multi-Dimensional Reinforcement Learning Using a Vector Q-Net - Application to Mobile Robots", in Proc. of 2002 FIRA Robot World Congress, pp.405-410, 2002.
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